

Application Program Development

Ch 01: Introduction to Computers and Programming

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Introduction to Computers and Programming

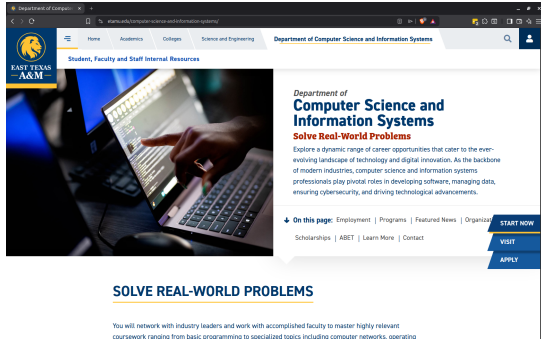


Figure 1: A web browser application program

- Computers can be programmed
 - Designed to do any job that a program tells them to
- **Program**: set of instructions that a computer follows to perform a task.
 - Commonly referred to as **Software**
- **Programmer**: person who can design, create and test computer programs
 - Also known as a **software developer**



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Computer Hardware Components

- **Hardware:** The physical devices that make up a computer
 - Computer is a system composed of several components that all work together

Typical major components:

- Central processing unit (CPU)
- Main memory (RAM)
- Secondary storage devices (hard drives, SSD)
- Input and output devices



Typical Components of a Computer System

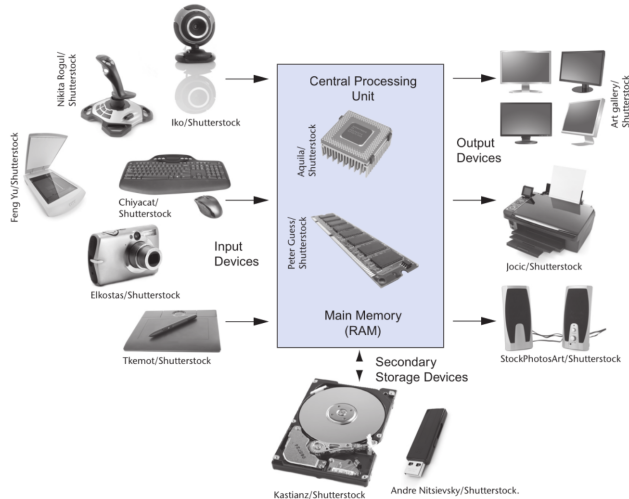


Figure 2: Typical components of a computer system (Gaddis, 2021, pg.25)

- Everything a computer does is controlled by software
- **Application software:** programs that make computer useful for every day tasks
 - Word processing, email, web browsers, programming editors, photo editing
- **System software:** programs that control and manage basic operations and resources of a computer
 - **Operating system:** controls operations of hardware components
 - **Utility programs:** perform specific tasks to enhance computer operation
 - **Software development tools:** used to create, modify and test software



How Computers Store Data

- All data in a computer is stored in sequences of 0s and 1s
- **Byte**: just enough memory to store letter or small number
 - Divided into eight bits
 - **Bit**: electrical component that can hold positive or negative charge, on/off switch
 - The on/off pattern of bits in a byte represents data stored in the byte

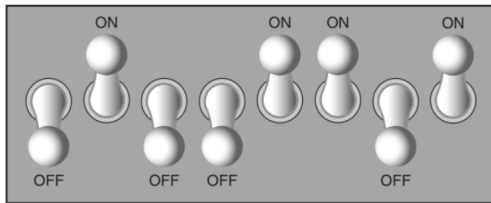


Figure 3: Think of a byte as eight switches (Gaddis, [2021](#), pg.30)



Storing Numbers

- Computers use binary numbering system
- $0100\ 1101 = 2^6 + 2^3 + 2^2 + 2^0 = 64 + 8 + 4 + 1 = 77$
- Byte size limits are 0 and 255
 - $0 = 0000\ 0000$ all bits off; $255 = 1111\ 1111$ = all bits on
 - To store larger number, use several bytes

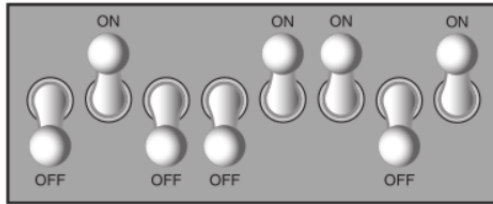


Figure 4: The number 77 stored in a byte (Gaddis, [2021](#), pg.30)



Storing Characters

- Data stored in computer must be stored as binary number
- Characters are converted to numeric code, interpreted as characters
 - Basic character encoding scheme is [ASCII](#) Appendix C (7 bits or 128 characters)
 - Unicode is current standard
 - Compatible with but extends ASCII
 - Can represent many character sets, for many other languages than English

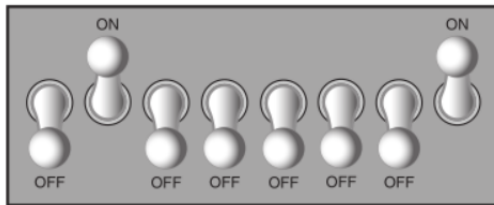


Figure 5: The letter A ASCII encoding in a byte (Gaddis, [2021](#), pg.30)



Advanced Number Storage

- To store negative numbers and real (float) numbers, computers use binary numbering encoding schemes
 - Negative signed numbers encoded using [two's complement](#)
 - Real numbers encoded using floating-point notation [IEEE 754](#)

Other Types of Data

- **Digital** describes any device that stores data as binary numbers
- Digital images are composed of pixels
 - Each pixel converted to a binary number representing pixel color
 - Video is a sequence of digital images
- Digital music and audio is composed of digitized sound samples
 - Each sound level sample is converted to a binary number



Summary

- An application program is a piece of software, created by programmers
- You should be familiar with the basic hardware components of a computer
 - CPU, RAM, secondary storage and I/O devices
- All data in a computer is stored as a sequence of 0s and 1s
 - A byte is a fundamental unit of computer data
 - Can use multiple bytes to represent larger numbers
- Not just numbers, can use different standards to digitally represent images, video, sound, and other types of data.



References

Gaddis, T. (2021). *Starting out with python* (5th). Pearson.



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